PROJECT DEVELOPMENT PHASE – 1

Development Phase Introduction

The EduGenie – Google Gemini Powered Learning Assistant is developed as an educational support application.

The development phase converts the approved project design and requirements into a working application.

The main focus is to build the user interface, connect the application with the Gemini-powered response system, process user questions, and display useful learning responses.

Development activities are performed module by module and checked continuously for errors.

Development Activities / Deliverables

• Development environment prepared

• Project files and folders organized

• Initial application structure created

Development environment / initial project structure

Paste the relevant project screenshot / output here.

Note: Replace the placeholder above with the actual screenshot/evidence from the EduGenie project before submission.

PROJECT DEVELOPMENT PHASE – 2

Development Environment & Project Setup

The required software environment is prepared before implementation.

The project source files are organized into suitable folders and modules.

The required programming language, development tools, libraries, and configuration files are prepared according to the project design.

Configuration values such as API credentials must be kept private and should not be exposed in screenshots or source repositories.

Development Activities / Deliverables

• Create project workspace

• Install required dependencies

• Configure application settings

• Run the initial project successfully

Project folder / IDE / setup screen

Paste the relevant project screenshot / output here.

Note: Replace the placeholder above with the actual screenshot/evidence from the EduGenie project before submission.

PROJECT DEVELOPMENT PHASE – 3

User Interface Development

The user interface is developed to provide a simple and student-friendly learning experience.

The interface includes an application title, question/input area, submit or ask action, and an area for displaying the generated response.

The layout is kept clear so that users can enter learning questions without confusion.

Basic validation is added to avoid processing empty or invalid inputs.

Development Activities / Deliverables

• Design input section

• Create response display section

• Add buttons and labels

• Check basic navigation and usability

EduGenie user interface screenshot

Paste the relevant project screenshot / output here.

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PROJECT DEVELOPMENT PHASE – 4

Gemini Integration & Question Processing

The core development task is connecting the learning assistant with the Gemini-powered response mechanism.

When a user submits a question, the application receives the input and prepares it for processing.

The request is sent through the configured integration, and the generated response is received by the application.

The response is then prepared for display in a readable format.

Development Activities / Deliverables

• Capture user question

• Validate and prepare input

• Send request to Gemini service

• Receive and display response

Gemini integration / response output screenshot

Paste the relevant project screenshot / output here.

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PROJECT DEVELOPMENT PHASE – 5

Core Module Development

The project is divided into logical modules so that development and maintenance are easier.

The input module handles user questions and basic validation.

The processing/integration module communicates with the Gemini service.

The response module displays the generated learning content to the user.

Error-handling logic is included so the application can provide a suitable message when a request cannot be processed.

Development Activities / Deliverables

• Input handling module

• AI response/integration module

• Output display module

• Error handling module

Source code / module structure screenshot

Paste the relevant project screenshot / output here.

Note: Replace the placeholder above with the actual screenshot/evidence from the EduGenie project before submission.

PROJECT DEVELOPMENT PHASE – 6

Testing During Development

Testing is performed continuously during development rather than waiting until the final stage.

Different types of educational questions are entered to check whether the application processes inputs correctly.

The team checks response display, empty-input handling, invalid input handling, and basic application flow.

Any errors found during development are corrected and the relevant module is tested again.

Development Activities / Deliverables

• Test normal question

• Test empty input

• Test unexpected input

• Verify response display

• Record and fix development errors

Test case / application output screenshot

Paste the relevant project screenshot / output here.

Note: Replace the placeholder above with the actual screenshot/evidence from the EduGenie project before submission.

PROJECT DEVELOPMENT PHASE – 7

Debugging, Security & Quality Improvements

Debugging is carried out to identify and correct coding or integration issues.

The application is checked for broken flows, incorrect output handling, and configuration problems.

Sensitive information such as API keys should not be hard-coded in publicly shared documents or repositories.

The interface and response formatting are reviewed to improve readability and consistency.

Only actual project screenshots and outputs should be inserted as evidence in the final submission.

Development Activities / Deliverables

• Debugging completed

• Sensitive credentials protected

• Output formatting checked

• Final development issues reviewed

Debugging / corrected output screenshot

Paste the relevant project screenshot / output here.

Note: Replace the placeholder above with the actual screenshot/evidence from the EduGenie project before submission.

PROJECT DEVELOPMENT PHASE – 8

Development Completion & Deliverables

The development phase results in a working version of the EduGenie learning assistant suitable for testing and demonstration.

The main interface, question processing, Gemini-powered response integration, output display, and basic error handling are completed according to the project scope.

Development evidence may include source-code screenshots, application screenshots, sample outputs, and testing records.

Before submission, all placeholder boxes in this document should be replaced with the team's actual screenshots or evidence.

The completed development build can then proceed to final testing, documentation, demonstration, and deployment activities.

Development Activities / Deliverables

• Working application build

• Source code

• Sample input/output evidence

• Development testing evidence

• Final project package

Final working EduGenie application screenshot

Paste the relevant project screenshot / output here.

Note: Replace the placeholder above with the actual screenshot/evidence from the EduGenie project before submission.